# Dengue Database — QA Report

File: **data/processed/dengue_cases_combined.xlsx** · Built by ml/clean_dengue.py · Report date: 18/10/2026

## Verdict: is it ready for EARS?

| | |
|---|---|
| **Overall** | **PASS** **Ready, with EARS settings.** Complete weekly calendar, nulls instead of gaps, a data_quality value on every row and a data_flag on suspect weeks. No data step is outstanding; what remains are settings in the EARS code (§5). |
| **NATIONAL sheet** | **PASS** **Ready.** 13 missing weeks and 14 year-end reporting artefacts are already flagged. EARS must skip any week whose data_quality is not “ok”. |
| **LOCAL sheet** | **WARN** **Ready, with flags.** 230 weeks of possible non-reporting (17 places) and 24 one-week spikes are marked in data_flag, values unchanged. The incomplete final week was removed. EARS must decide how to treat flagged weeks and use floors for small counts (§5). |

## 1. Dataset overview

| | NATIONAL | LOCAL |
|---|---|---|
| Rows | 261 | 32,886 |
| Series | 1 (Philippines) | 126 provinces / cities in 17 regions |
| Weeks | 261 (10/01/2016 – 03/01/2021) | 261 per place (same range) |
| Source | OpenDengue V1.3 (WHO WPRO / MOH) | DOH local-level dataset (HDX) |
| data_quality | ok 234, missing 13, incomplete 8, backlog 6 | ok 31,910, suspect swap 665, missing 311 |
| data_flag | none | possible non-reporting 230, one-week spike 24 |
| doh_status | no baseline 167, normal 49, epidemic 40, unreliable week 4, alert 1 | no baseline 19,537, normal 8,839, epidemic 2,995, alert 1,113, unreliable week 402 |

Columns: geographic_level, (region, geography_name — local only), period_start, value, data_quality, data_flag, epi_year, epi_week, alert_threshold, epidemic_threshold, doh_status, doh_note.

## 2. Checks passed

| Check | Result | |
|---|---|---|
| No duplicate place-week rows | 0 duplicates | **PASS** |
| Every date is a Sunday (week start) | all 33,147 rows | **PASS** |
| Complete weekly calendar, no skipped weeks | 126 × 261 local, 261 national | **PASS** |
| Each place belongs to one region | 0 conflicts | **PASS** |
| Values are non-negative whole numbers | 0 negative, 0 non-integer | **PASS** |
| Null value ⇔ data_quality = missing | consistent on both sheets | **PASS** |
| Epi weeks numbered 1–52 (53 in 2020) | matches the MMWR calendar | **PASS** |
| DOH thresholds recomputed independently | 11,436 rows, 0 mismatches | **PASS** |
| data_flag only on reported (“ok”) weeks | 0 on non-ok rows | **PASS** |
| No blank doh_status cells | every row has a status + note where needed | **PASS** |
| National vs local totals agree | correlation 0.95; national ≈ 0.92× local sum (191 complete weeks) | **PASS** |

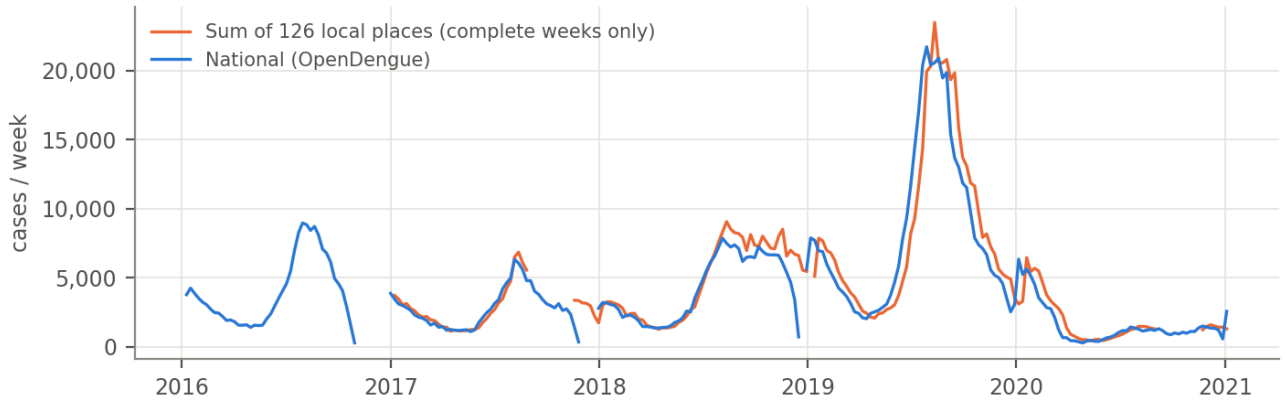



Figure 1. National series against the sum of all local places. They track closely except at year-end (national fades out, then jumps) and during the 2019 epidemic peak. The local sum is drawn only for the 196 weeks where all 126 places have a valid value: 15 weeks with suspected cases/deaths swap rows and 55 weeks with missing rows have no total, rather than a partial sum that would look like a dip. The excluded rows stay in the workbook, marked by data_quality.

## 3. Findings

Ranked by how much they affect EARS. The status after the latest rebuild is shown on each finding.

### HIGH F1. Local zeros that look like non-reporting — flagged

230 weeks in 17 places are runs of 4+ consecutive zeros in a place that normally reports 10+ cases a week. These are almost certainly “no report received”, recorded as 0 — the source does not distinguish the two. They cluster in recognisable events: **BARMM in 2017** (Lanao del Sur and Maguindanao during the Marawi siege), **Apr–Jun 2020** (COVID-19 lockdowns: NCR cities, Baguio, Angeles, Zambales), **Tacloban Jan–Sep 2016**, and **Eastern Visayas Nov 2020 – Jan 2021** (end of the series).

**Can it be fixed?** Not from this data: the true counts for these weeks were never recorded. Only DOH / the regional surveillance units hold corrected figures, so this is a reporting-system limitation rather than a cleaning error.

**Why it matters:** a zero inside the 7-week baseline drags the mean and SD down, so the first real week after the run looks like an outbreak. **Done:** these weeks have data_flag = “possible non-reporting” (rule: 4+ consecutive zeros where the place’s median non-zero week is 10+). Values are kept as published; doh_note repeats the flag.

| Place | Region | Zero run | Weeks | Typical / wk |
|---|---|---|---|---|
| TACLOBAN | REGION VIII-EASTERN VISAYAS | 10/01/2016 – 25/09/2016 | 38 | 10 |
| MAGUINDANAO | BARMM | 08/01/2017 – 14/05/2017 | 19 | 17 |
| MAGUINDANAO | BARMM | 27/08/2017 – 19/11/2017 | 13 | 17 |
| MAGUINDANAO | BARMM | 28/05/2017 – 13/08/2017 | 12 | 17 |
| NORTHERN SAMAR | REGION VIII-EASTERN VISAYAS | 01/11/2020 – 03/01/2021 | 10 | 10 |
| LANAO DEL SUR | BARMM | 05/11/2017 – 31/12/2017 | 9 | 16 |
| TACLOBAN | REGION VIII-EASTERN VISAYAS | 09/10/2016 – 27/11/2016 | 8 | 10 |
| LANAO DEL SUR | BARMM | 08/01/2017 – 26/02/2017 | 8 | 16 |
| ZAMBALES | REGION III-CENTRAL LUZON | 05/04/2020 – 17/05/2020 | 7 | 13 |
| TAGUIG CITY | NATIONAL CAPITAL REGION | 12/04/2020 – 24/05/2020 | 7 | 15 |
| LANAO DEL SUR | BARMM | 09/07/2017 – 20/08/2017 | 7 | 16 |
| BAGUIO CITY | CAR | 12/04/2020 – 24/05/2020 | 7 | 10 |

Top 12 of 30 runs shown.

### HIGH F2. Local one-week spikes — flagged

24 weeks are more than 8× the place’s 9-week rolling median and at least 4× both of their neighbours: the count jumps for a single week and falls straight back (e.g. Cebu: 0 → 2,123 → 1 in Sep 2017). Real outbreaks build and decline over several weeks, so these are most likely backlogs dumped into one week or entry

errors. Even the spikes inside the 2019 epidemic (Zamboanga City 298 → 3,382 → 377, Antique 209 → 2,036 → 196) have this shape. A short manual review against DOH bulletins is advised.

**Can it be fixed?** Not reliably: there is no record of which earlier weeks the cases belong to, so spreading them back would be a guess. **Done:** these weeks have data_flag = “one-week spike”, values kept. If a review against DOH bulletins confirms one as real, its flag can be removed.

| Place | Week | Week before | Cases | Week after | 9-wk median | × larger neighbour |
|---|---|---|---|---|---|---|
| ZAMBOANGA CITY | 28/07/2019 | 298 | 3,382 | 377 | 280 | 9.0× |
| ANTIQUE | 11/08/2019 | 209 | 2,036 | 196 | 196 | 9.7× |
| CAGAYAN DE ORO CITY | 23/09/2018 | 104 | 998 | 131 | 109 | 7.6× |
| BOHOL | 11/09/2016 | 125 | 934 | 83 | 86 | 7.5× |
| CAGAYAN DE ORO CITY | 18/11/2018 | 94 | 818 | 94 | 94 | 8.7× |
| LANAO DEL NORTE | 11/11/2018 | 73 | 686 | 54 | 66 | 9.4× |
| ANTIQUE | 13/08/2017 | 36 | 678 | 72 | 30 | 9.4× |
| CEBU CITY | 06/08/2017 | 60 | 673 | 59 | 57 | 11.2× |
| ILOCOS NORTE | 10/11/2019 | 78 | 652 | 44 | 78 | 8.4× |
| LA UNION | 02/12/2018 | 24 | 551 | 33 | 30 | 16.7× |
| NEGROS ORIENTAL | 06/08/2017 | 58 | 546 | 62 | 60 | 8.8× |
| BUKIDNON | 06/08/2017 | 46 | 542 | 60 | 57 | 9.0× |
| LA UNION | 18/11/2018 | 30 | 335 | 24 | 35 | 11.2× |
| OCCIDENTAL MINDORO | 08/07/2018 | 13 | 333 | 39 | 35 | 8.5× |
| PUERTO PRINCESA CITY | 30/06/2019 | 19 | 332 | 42 | 19 | 7.9× |
| MANDAUE CITY | 17/02/2019 | 42 | 322 | 28 | 28 | 7.7× |
| TANDAG CITY | 19/03/2017 | 5 | 229 | 3 | 3 | 45.8× |
| ZAMBALES | 10/11/2019 | 15 | 221 | 17 | 20 | 13.0× |
| ISABELA | 09/02/2020 | 29 | 199 | 11 | 14 | 6.9× |
| ROMBLON | 17/11/2019 | 4 | 121 | 0 | 4 | 30.2× |
| PATEROS | 19/03/2017 | 1 | 94 | 0 | 0 | 94.0× |
| PATEROS | 04/06/2017 | 0 | 86 | 0 | 0 | 86.0× |
| DAVAO DEL NORTE | 30/07/2017 | 7 | 67 | 1 | 4 | 9.6× |
| LAPU-LAPU CITY | 15/05/2016 | 2 | 36 | 1 | 2 | 18.0× |

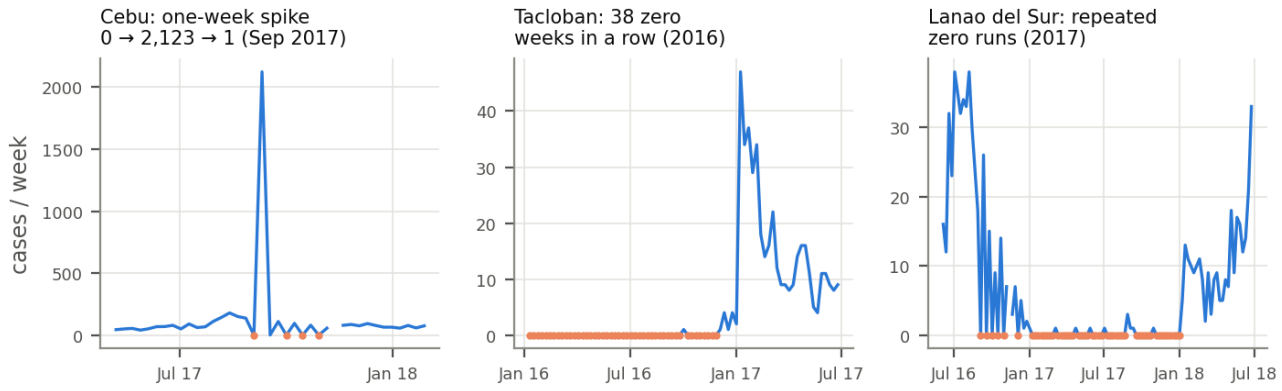



Figure 2. Examples of F1 and F2. Orange dots are zero weeks.

### MEDIUM F3. Local final week was incomplete — removed

The local data’s last week (10/01/2021) totalled 784 cases against about 1,384 in the weeks before: it was extracted before most of its reports had arrived. **Done:** both sheets now end on 03/01/2021, so national and local still cover the same 261 weeks.

### MEDIUM F4. Small counts make thresholds hair-trigger

Places with very few cases produce baselines with little or no spread: 121 rows have a baseline SD of 0 (alert = epidemic threshold, so any case above the mean is an “epidemic”), and 157 local weeks are marked “epidemic” with 5 cases or fewer — mostly Davao Occidental, Tawi-Tawi, Sulu, Catanduanes and Dinagat Islands. 8 places are zero in over a quarter of weeks (Batanes 69%).

**Not a data error** — this is how the data is. It is handled in the EARS settings (§5): a minimum SD and a minimum case count before an alarm is raised.

### LOW F5. Davao Occidental is patchy

54 missing weeks scattered through 2016–2018 and an average of 1 case a week. EARS will rarely have 5 usable baseline weeks there before 2019; expect mostly blank results.

### INFO F6. Cities and provinces listed side by side

Highly urbanised cities appear as their own rows (e.g. CEBU and CEBU CITY). The local sum is close to the national total, which suggests provinces exclude their cities, but this is not documented. Do not add a city to its province without confirming.

### INFO F7. National year-end artefacts — already handled

Each year national fades out before a December gap (“incomplete”, 8 weeks) and jumps in early January (“backlog”, 6 weeks). These are flagged and excluded from DOH thresholds.

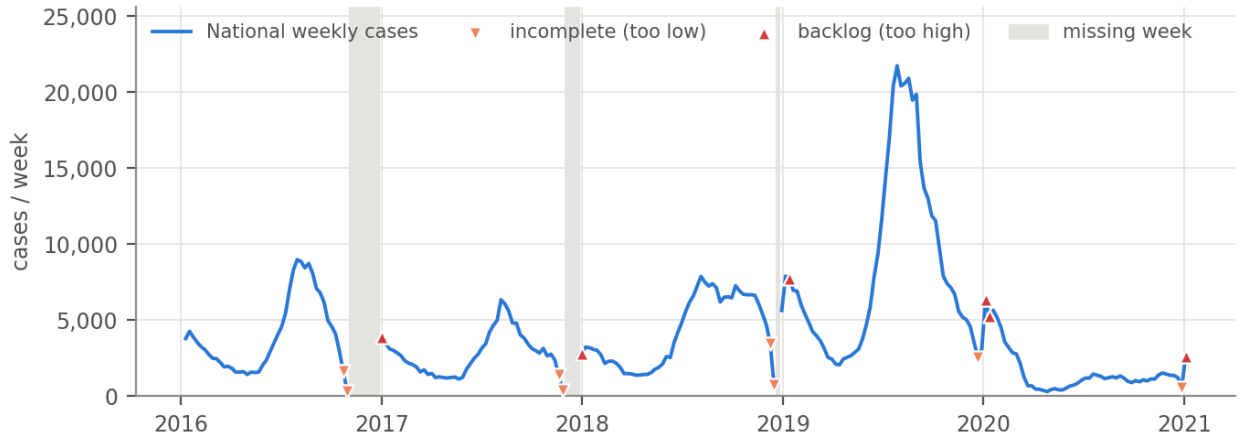



Figure 3. National series with its data_quality flags. Grey bands are missing weeks.

## 4. Cleaning already applied

| Step | Detail |
|---|---|
| Trimmed to a common range | 10/01/2016 – 03/01/2021: the local data’s span minus its incomplete final week; national 2012–2015 and 2021–2023 dropped |
| Complete weekly calendar | every series has all 261 weeks; missing weeks kept as null rows, never deleted, zero-filled or interpolated |
| Dropped columns | national: everything except geographic_level, period_start, value; local: deaths |
| Removed non-data rows | national title/description rows; local HXL tag row (#adm2+name …) |
| Name fixes | BASILIG CITY → BISLIG CITY, NUEVA VISCAYA → NUEVA VIZCAYA, SISQUIJOR → SIQUIJOR; leading spaces trimmed |
| Region fixes | REGION VII-EASTERN VISAYAS → REGION VIII-EASTERN VISAYAS; REGION IVB → IV-B; case unified |
| data_quality flag | ok / missing on both sheets; incomplete / backlog on national year-end weeks |
| data_flag | local: “possible non-reporting” (F1) and “one-week spike” (F2); values unchanged |
| DOH thresholds | Per DOH PIDSR Manual of Procedures (2014) §5.3.5, over the same epi week in 3 prior years: alert = mean + 1·SD, epidemic = mean + 2·SD (sample SD); 2-of-3 fallback; week 53 uses week 52; every deviation explained in doh_note |

## 5. EARS readiness checklist

| Requirement | National | Local |
|---|---|---|
| One row per week, no skipped weeks | **PASS** | **PASS** |
| Missing weeks are null, not deleted or zero | **PASS** | **PASS** |
| Unreliable weeks flagged so baselines can skip them | **PASS** | **PASS** — F1, F2 in data_flag; F3 removed |
| Enough history for a 7-week baseline (C1–C3) | **PASS** | **PASS** (except Davao Occidental) |
| Counts large enough for a stable SD | **PASS** | **WARN** — F4, set floors |

### Before running EARS

**Data:** nothing outstanding. Optional: review the 26 F2 spikes against DOH bulletins.

**EARS settings (in the EARS code):**

1. Use only weeks with data_quality = “ok”; any other week is treated as null.
2. Weeks with a data_flag still get their own EARS result, with the flag repeated in the note. Leave them out of other weeks’ baselines (recommended), or keep them in and note every result whose baseline contains one.
3. Compute a baseline only when at least 5 of its 7 weeks are usable (the agreed middle-ground rule); otherwise no result.
4. No result when the current week is not “ok”; C3 needs C2 at t, t−1 and t−2.
5. Set a minimum SD (e.g. 1 case) so a flat baseline cannot divide by zero, and require a minimum count (e.g. 5 cases) before a week can alarm, to stop single-case alarms in small places (F4).

Limitation for the paper: F1 and F2 reflect late or missing reporting at source. EARS can only be as timely and complete as the reports it receives.